Though execution time is a bit longer compared to Java, Ruby works here too. For a more sophisticated version of your application, it's advisable to put it into a single file. But first, let's make our Ruby experience a bit more pleasant:

```
$ cp  IRB-0.5.2-release.apk  ruboto.jar
$ dalvikvm -classpath  ruboto.jar org.jruby.Main -X-C -e "require
'java'; puts java.lang.System.get_property('java.vendor')"
The Android Project
```

Well, Ruby operations work well. Now let's put our program into a separate file, named *test.rb:*

```
require 'java'
import java.lang.System

class Ruboto
def greet(who)
  puts "Hello, #{who}!"
end
end

name = System.get_property('java.runtime.name')
Ruboto.new.greet(name)
```

As you can see, our Ruby script retrieves a Java environment property. Run it thus:

```
$ dalvikvm -classpath ruboto.jar org.jruby.Main -X-C test.rb
Hello, Android Runtime!
```

Getting Python installed

If you want Python-based applications too, you'll have to transfer a Python installer onto the phone, and install the full Python environment from it. Transfer the installer via SSH (Wi-Fi or USB connection) or simply run an *adb* command like the following, on your PC:

```
$ ./platform-tools/adb install -s /home/anton/python_for_android_
r1.apk
774 KB/s (31714 bytes in 0.039s)
        pkg: /sdcard/tmp/python_for_android_r1.apk
Success
```

Now, go to the main screen and click the Python icon to launch it, in order to install all other modules. After a while, installation is complete, and you can run the interpreter. Please pay attention to where Python hides: */data/data/com.googlecode.pythonforandroid/files/python/*. Thus, to run Python from the command-line, use the following code:

```
$ /data/data/com.googlecode.pythonforandroid/files/python/bin/
python
Could not find platform independent libraries <prefix>
Consider setting $PYTHONHOME to <prefix>[:<exec_prefix>]
'import site' failed; use -v for traceback
Python 2.6.2 (r262:71600, Sep 19 2009, 11:03:28)
[GCC 4.2.1] on linux2
Type "help", "copyright", "credits" or "license" for more
information.
>>> print "Hello, World!"
Hello, World!
```

Again, it's advisable to put your app into a single script file, as well as set up the necessary variable, namely *$PYTHONHOME.*

In addition to these languages, Google also offers Perl, Lua [see Reference 7] and even more. As you can see, Android is a fun thing—attach a big keyboard and screen, and you'll get a developer's machine for Android, under Android :)

Android smartphones are everywhere these days. You can find interesting models like the Acer beTouch E110, the LG Optimus GT540, or the Samsung Galaxy Next. The same companies offer dual-SIM models as well, like the LG GX500 and Samsung GT-B7722 Duos. However, neither of the two qualify as an Android-based device—and, to be honest, the prices for the dual-SIM models are almost twice those of a single-SIM unit! With this in mind, devices based on the MT6516 look very attractive, especially when you consider their price, quality and technical capabilities. And then, don't forget one aspect—this isn't simply an open source Android platform, but a full-featured (well, almost full-featured) Linux computer. It can run Java applications, Ruby and Python (and other) scripts, or even ordinary binaries for the ARM processor. Isn't it a good starting point for newbies, and even professionals, to start learning something new about Android, and Linux in general? All you need is a device, your skills and knowledge, and a will to program. END

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